

Hamza Shaikh

Site Reliability · DevOps · Cloud Infrastructure

er.hamzashaikh@gmail.com | linkedin.com/in/shamza31 | github.com/Raynzler | shamza31.vercel.app | Germany

Education

M.Sc. Computer Science, RPTU Kaiserslautern-Landau, Germany *Oct. 2025 – Expected 2027*

Specialisation 1: Distributed Systems | Specialisation 2: Software Engineering

B.E. Information Technology, Xavier Institute of Engineering, Mumbai, India *Aug. 2020 – Jun. 2024*

CGPA **8.69 / 10** | *Career break July 2024 – Sep. 2025: M.Sc. admissions, intake deferred to Oct. 2025*

Technical Skills

Languages: Python, Go, C++, Bash, SQL

Cloud & Infrastructure: AWS (EC2, ECS, Fargate, VPC, Subnets, Security Groups, IAM, S3, CloudWatch, ELB), Docker, Docker Compose, Linux, Terraform, NGINX, Cloudflare, DigitalOcean, Kubernetes (foundational)

Observability & Reliability: Prometheus, PromQL, Grafana, Alertmanager, RED metrics, SLOs, chaos engineering, incident response

CI/CD & Security: Git, GitHub Actions, pytest, Bandit, Trivy, pip-audit, govulncheck, container hardening

Data & Services: FastAPI, Redis, BullMQ, MySQL

Experience

DevOps Engineering Intern (Sole Infrastructure Owner)

Jun. 2023 – Aug. 2023

Patil Kaki, D2C Food Brand | Remote

- Migrated production from EC2 to ECS Fargate as the only infrastructure engineer in a 10-person team, supporting a storefront at ~2,500 orders/month; dynamic task provisioning cut AWS spend 27% (\$230 to \$168/month).
- Built the AWS network and access layer: VPC subnets, routing and security groups, S3 storage, and least-privilege IAM roles scoped separately for founders, developers and logistics staff.
- Root-caused a three-day chatbot outage: a whitespace character intermittently injected into a hosting-generated integration URL was silently breaking the widget, with no alert and no logged error.
- Built GitHub Actions pipelines gating main-branch deployments with rollback; decoupled order processing from the request path using BullMQ over Redis.

Projects

AutoSRE: Chaos Engineering Platform | *Python, Go, FastAPI, Prometheus, Docker*

Feb. – Jul. 2026

- Built a chaos engineering framework injecting latency, errors, CPU burn and memory pressure into live services through a REST control plane. Experiments are bounded and self-reverting; the control plane is exempt from injection so the abort path can never be made unreachable.
- Implemented a three-state circuit breaker (closed/open/half-open) with an injectable clock for deterministic tests and Prometheus state metrics, plus token-bucket rate limiting middleware. 93.9% line coverage on the shared library.
- Instrumented RED metrics via ASGI middleware with bounded label cardinality; defined a p95 500ms SLO as Prometheus rules. CI runs Bandit, Trivy, pip-audit and govulncheck with actions pinned to commit SHAs; containers run read-only and non-root.

SentinelSQL: Distributed Node Monitoring | *Go, Prometheus, Alertmanager*

Sep. 2025 – May 2026

- Built a Go telemetry daemon polling a Solana validator's JSON-RPC out of process into Prometheus; backoff retries only transient failures, and both endpoints scrape concurrently under a WaitGroup so the metrics share a timestamp.
- Implemented 3-sigma Z-score anomaly detection as Prometheus recording and alerting rules against a rolling baseline learned from the node's own history, routed through Alertmanager.
- Fixed a detection blind spot: the original rate-ratio rule went undefined as the validator stalled, silencing the alert at exactly the moment it existed to catch. **3rd Place**, Superteam Germany Ideathon, Frankfurt (2026).

GridCast: Heatwave Forecasting | *Python, TensorFlow, LSTM, Flask, React*

Aug. 2023 – May 2024

- Built 28 region- and season-specific LSTM models on 70 years of gridded temperature data (1951–2021), served through a Flask API behind a React and D3 frontend. B.E. capstone project.

Additional

Languages: English (professional), Hindi, Urdu, Arabic (conversational), German (basic)

Certification: AWS Certified Cloud Practitioner (2023)

Community: Hack The League (Mumbai), Hack This Fall (Pune) 2022; Solana Hacker House BLR/MUM 2023, BLR 2024; Berlin Blockchain Week, Solana Summit Germany 2026, ETH Day Berlin 2026

Work authorisation: Indian national, German student residence permit. Requires sponsorship outside Germany, India and Saudi Arabia. Available from Oct. 2026.